

Cirrhosis of the liver is characterized by hepatic cell death, tissue scarring, and altered liver function. All of the following would be disrupted by cirrhosis of the liver EXCEPT:

- ☐ A. the production of ketone bodies. [8%]
- ☐ B. the metabolism of most drugs. [2%]
- ☐ C. the oncotic pressure within capillaries. [46%]
- ✔ ☒ D. the secretion of digestive proteolytic enzymes. [43%]

Correct

43% answered correctly

Explanation:

Function of liver cells
• Regulation of blood glucose via glycogenesis, glycogenolysis & gluconeogenesis • Storage of glycogen, minerals (iron) & vitamins • Synthesis of macromolecules such as plasma proteins (clotting factors & albumin), fats, ketone bodies & cholesterol • Production & secretion of bile • Breakdown/detoxification of numerous drugs & metabolic waste products (bilirubin, ammonia)

The **liver** is located in the upper right side of the abdominal cavity, just below the diaphragm, and performs various functions within the body. According to the question, cirrhosis causes hepatic cell death and scarring. As a result, the following *normal* functions of the liver are severely impacted:

- **Ketogenesis:** Ketogenesis (synthesis of ketone bodies) occurs mainly in the liver and plays a critical role in energy metabolism when glycogen stores are depleted and blood glucose levels become low. During times of fasting, mitochondria within liver cells convert acetyl coenzyme A into ketone bodies, which are then used by the brain to make ATP (**Choice A**).
- **Detoxification of drugs:** Drugs are often exogenous compounds that may have toxic effects if allowed to accumulate in the body. The liver is a critical organ for the breakdown and detoxification of many drugs (**Choice B**).
- **Synthesis of plasma proteins:** Osmotic pressure within capillaries is due to the presence of plasma proteins that cannot easily cross the capillary membrane. Oncotic pressure causes a "pulling force" inside capillaries, which balances the pushing force of hydrostatic pressure, helping keep fluid in the vasculature. The majority of plasma proteins such as albumin are synthesized in the liver. Therefore, cirrhosis is expected to impact the production of plasma proteins, which in turn would affect oncotic pressure in the capillaries (**Choice C**).

(**Choice D**) **Digestive proteolytic enzymes**, which hydrolyze polypeptides, are synthesized and secreted from the pancreas, stomach, and small intestine. Cirrhosis of the liver is not expected to affect the secretion of digestive proteolytic enzymes because they are secreted by other tissues.

Educational objective:

The liver has various functions, including storage of glycogen, regulation of blood glucose levels, detoxification of drugs, synthesis and secretion of bile, production of serum proteins (eg, albumin, clotting factors), and synthesis of ketone bodies.

Time spent: 0 sec
Subject:
Biology

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System:
Digestion and Excretion